

Sen Zhang

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Summary

Registered NY architect (NCARB) and computational design technologist, 9 years post-Master's. Single-person Design Technology department for a 150-person firm: architect and lead developer of EnneadTab — a 450+ tool plugin platform with three U.S. Copyright registrations — plus BimRunner, EnneadTabLLM, FieldDay, and AIA-Sustainable-Reporting. Independent product work spans real-estate networking platforms, social-media small-business apps, and a cross-machine agentic Claude ecosystem. Speaker: Autodesk University 2024, Advancing Computational Design 2026.

Experience

Ennead Architects LLP / Computational BIM Developer

New York, NY · 2022–Present

BIM Manager (concurrent)

2022–Present

Rhino Doctor (concurrent)

2022–Present

Architectural Designer, BIM Specialist

2020–2022

- **Computational BIM Developer — single-person Design Technology department for a 150-person firm**

- + Architect and lead developer of EnneadTab — 450+ tools across Revit, Rhino, CAD, Adobe, Excel, Email, and LLM integration; three U.S. Copyright registrations (2024)
- + BimRunner — Revit project-health analytics dashboard (Next.js 14 + Vercel Postgres, ~91k LOC, 422 files); RESTful ingestion + dynamic scoring
- + EnneadTabLLM — on-prem Ollama RAG monorepo for HIPAA / data-sovereign clients
- + AIA-Sustainable-Reporting — CRON Excel pipeline + dashboard for AIA 2030 metrics; FieldDay — offline-first PWA for site inspection with PDF spatial pinning
- + EnneaDuck — pixel-mascot community-engagement system that gamifies internal adoption of design-tech across 150 staff
- + Fluent in CPython, IronPython, Revit API, RhinoCommon API, TypeScript / Next.js; LLM integration via Ollama, OpenAI, Anthropic, MCP

- **BIM Manager (concurrent)**

- + Healthcare (NE US): NYU Langone, Northwell, Lenox Hill
- + Commercial (Asia): FTZ Shanghai, ByteDance Houhai HQ, Bilibili HQ
- + Solution architect for large-model BIM data, complex families, in-house training and mentoring

- **Rhino Doctor (concurrent)**

- + Rhino-to-Revit interoperability via zero-learning-curve internal tooling
- + Triage of complex parametric and industrial-scale geometry
- + Speaker: Autodesk University 2024 (San Diego), Advancing Computational Design Conference 2026 (Austin)

Arquitectonica / Designer, BIM Specialist, CG Visualization

New York, NY · 2017–2020

- 40 Hudson Yards (NYC office tower); Amazon HQ2 master planning (Chicago); 31 Parliament mid-rise residential tower (Toronto SmartCity)
- Curtis Block (Calgary); Warsaw Hotel (Poland); Newport Waterfront; Equinox Hotel (San Jose); Queens Sherbourne mixed-use with historical preservation (Toronto)

RPI Publication Dept. / Book-design, Asset Management

Troy, NY · 2016–2017

Terreform ONE / Research, Design

New York, NY · 2016–2017

Selected Projects

Firm Tools — Ennead Architects

EnneadTab / Architect & Lead Developer

2021–Present

Flywheel-ecosystem plugin platform spanning Revit, Rhino, CAD, Autodesk Desktop Connector, Enscape, InDesign, Photoshop, Excel, Email, AI/LLM, computer-resource monitoring, text-to-speech, and sound effects. 500+ tools across the family of products, three U.S. Copyright registrations (2024). 63,627 logged usage events in Q4 2025 alone across 60 active daily users — averaging 17x/person/workday. CPython compile pipeline → distribution repo → silent user-side scheduled updater with delayed-deletion background swaps. "Best design of innovation is invisible."

BimRunner / Architect & Full-Stack Developer

2026–Present

Production Revit project-health analytics dashboard. Next.js 14 + Vercel Postgres + Chart.js, ~91k LOC across 422 files. RESTful ingestion API, dynamic scoring from stored facts, file-attachment extensibility. Surfaces model debt before it becomes coordination crisis.

EnneadTabLLM / Architect & Developer

2025–2026

Local-deployment Ollama RAG monorepo paired with the EnneadTab Wiki — what the Austin 2026 talk frames as "Cloning the Expert." Snowballing knowledge base ingests years of standards, templates, family libraries, and chat-history troubleshooting; surfaces curated tool chains across 500+ EnneadTab commands. Windows server + Vercel Next.js proxy + shared TS types; solves the on-prem AI problem for HIPAA / data-sovereign clients without sending model data to a vendor.

AIA-Sustainable-Reporting / Architect & Lead Developer

2026–Present

CRON Excel pipeline + frontend dashboard for AIA 2030 sustainable-design metrics. Single-source `.env.local` config across Next.js app, cron scripts, and CLI; Vercel CLI + Codespaces friendly. Closes the firm-wide reporting loop for the AIA 2030 commitment.

FieldDay / Architect & Lead Developer

2026–Present

Offline-first PWA for site inspection on phones, tablets, and desktops. PDF-based spatial pinning, photos, voice notes, and comments with zero-connectivity capture and Vercel Pro cloud sync. Preserves spatial accuracy across drawing revisions.

RenderPolisher / Architect & Developer

2025–2026

AI-driven render post-processing pipeline. Cleans and uplifts architectural visualizations through a configurable model chain; integrates into the Ennead delivery workflow via the EnneadTab toolbar.

EnneadTabHome / Architect & Developer

2025–2026

Landing page and central hub for the EnneadTab application family at enneadtab.com. Next.js 14, Tailwind, mobile-friendly UI.

EnneadTabWiki / Architect & Lead Author

Ongoing

Public documentation site for the EnneadTab plugin ecosystem. Auto-generated from source repos; serves as both internal training material and external industry signal. <https://ennead-architects-llp.github.io/EnneadTabWiki/>

EnneaDuck / Designer & Developer

2022–Present

Pixel-mascot community-engagement system embedded throughout EnneadTab — randomized achievements, status notifications, and gamified adoption hooks that turned a top-down tooling rollout into a 150-person voluntary user base.

Rosette / Architect & Developer

2026–Present

Architectural-language translator: turns design-team, client, and regulatory text into one-line plain-English (or Mandarin) summaries with an interactive structural graph. Part of the EnneadTab product line.

2025 Performance Review / Architect & Lead Developer

2025–2026

Firm-wide employee-evaluation system. Excel ingest → interactive HTML report with software-proficiency tracking and visual analytics. Deployed to GitHub Pages.

Independent — Founder / Engineer

Realm / Founder & Engineer

2025–Present

Networking and event platform for architects, real-estate operators, and exhibition partners. Next.js 16 + NextAuth v5 + Drizzle ORM + Vercel Postgres. Multi-step RSVP, admin/client dashboard, password-reset flows via Resend, role-based partner gating.
<https://realmnomadgroup.com>

Hair-Salon SMB Platform / Founder & Engineer

Ongoing

End-to-end booking, inventory, and customer-loyalty platform for a NYC hair salon — small-business operator using AI workflows to replace Square + spreadsheets + Instagram DMs with a single owner-facing dashboard.

Social-Media Ads Service / Founder & Engineer

Ongoing

Ad-management service for small-business social-media campaigns. Pipelines creative-brief intake, asset generation, scheduling, and performance reporting through a Claude-orchestrated agent team.

Social-Media Channel — Android App / Founder & Engineer

Ongoing

Native Android application powering a community social-media channel. Custom feed pipeline, content moderation, and creator-side analytics built end-to-end on top of the same agentic infrastructure that drives the firm-side tooling.

LakeHouse-Design.com / Architect & Developer

Ongoing

Rhino package of professional design tools — dynamic area takeoff, layer/material merging, object transformation, block management.
<https://lakehouse-design.com/>

AI Ecosystem — Cross-machine Agentic Infrastructure

Cross-Machine Agentic Claude Ecosystem / Architect & Developer

2025–Present

End-to-end portable Claude Code configuration — settings, skills, MCP servers, custom commands, plugins, project memories, and plans — that synchronizes across Mac, Windows, and mobile so the same agent team is available everywhere. 1,700+ commits in 90 days. Personal application of the "agentize humans" thesis from the Austin 2026 talk: AI is part of every operator's mind extension, not a chat window.

ideaFactory / Architect & Developer

2025–2026

Virtual-company multi-agent pipeline as a Claude Code plugin: each role agent runs independently and collaboratively, daily, reporting to a Project-Manager orchestrator. Modernized to Gemini Imagen 4.0 + TTS.

youtube-plus / Architect & Developer

2026

MCP server exposing YouTube as a programmable tool surface for Claude — search, transcript extraction, comment analysis, channel exploration, analytics, and upload — used inside the agent ecosystem.

senzhang-wechat-harness / Architect & Developer

2026

macOS desktop-automation harness that lets Claude read and reply to WeChat messages via AppleScript + screencapture — no API key, works for any WeChat account on a connected Mac.

Speaking & Recognition

Advancing Computational Design Conference / Speaker

Austin, TX · 2026

The Design of Design — Enabling Agile Introduction of New Tools to Scale Design Efficiency <https://2026-ai-talk.vercel.app/>

Autodesk University / Session Speaker

San Diego, CA · 2024

Revit As A Game Engine

Digital Built Week Americas / Panel Speaker

Dallas, TX · 2023

Promoting Computational Design to Non-Programmers

AEC Hackathon / Team — Honorable Mention

New York, NY · 2023

Educational Tool Connecting Junior and Senior Students

Education

Columbia University GSAPP / M.S. Advanced Architecture Design	New York, NY · 2016–2017
Rensselaer Polytechnic Institute / Bachelor of Architecture	Troy, NY · 2012–2016
Tongji University / Major in Urban Planning	Shanghai · 2010–2012

Credentials & Technical Stack

• Registered Architect, New York (NCARB)	
• Languages: Python (CPython, IronPython), TypeScript, React, Next.js	
• Tooling: Revit API, RhinoCommon API, Grasshopper, Dynamo, MCP, LLM integration (Ollama, OpenAI, Anthropic)	
• Spoken: English, Mandarin Chinese (native)	

Awards & Recognition

• EnneadTab — U.S. Copyright Office — 3 registrations, authorship credited to Sen Zhang	2024
• Hyperloop Campus, Las Vegas — Honorable Mention	2020
• Greenway to Ideal City, Chengdu — 2nd Place	2019
• Tokyo Anti-Library — Honorable Mention	2018
• Non Architecture: Dance — Finalist	2018
• AIA Henry Adams Award	2016
• Magna Cum Laude Honors — Dean's List 2012–2016	2016
• George T. Droste Scholarship — Structural engineering study	2015
• Faculty Selection Award — Academic proficiency in Architecture	2015